

From demo to production

Attendee guide. Take Atlas, an agent that runs on your laptop, and put it on Cloud Run: a container, a private deploy, autoscaling, a load test, and chats that survive in Firestore.

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Built with the Google Agent Development Kit, Gemini, Docker and Cloud Run

Written for complete beginners: it starts from an empty laptop

Every screenshot and every terminal block in this guide comes from a real run.

Long output is shortened, and the project ID is shown as `my-atlas-123456`

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1 What you will do

You shipped an AI product in four weeks. It works on your laptop. Now what happens when a hundred people open it at once?

In this session you take one working agent and move it to Google Cloud without rewriting it. The agent is Atlas, a map agent: you tell it where you are, and it finds three places worth the walk, describes each one from checked sources, and draws the route. You do not need to have attended the session where it was built. The finished app is in the repository.

STEP	WHAT YOU DO	WHAT YOU LEARN
1	Put the app in a container	What a Dockerfile is, line by line
2	Run the container on your laptop	The same image runs anywhere
3	Push the image to Google Cloud	What a registry is for
4	Deploy it to Cloud Run, privately	One command, and who is allowed in
5	Read the logs	How to see what your app did in the cloud
6	Send 20 users at once	One instance takes them all
7	Tune concurrency	Cloud Run starts more instances, and the app starts forgetting people
8	Store chats in Firestore	State that survives across instances

This is what you end up with:



Docker: the app in a box | Cloud Run: runs the box and copies it under load | Firestore: memory all copies share

2 Before you start

You do not need to have used Docker or Google Cloud before. This guide starts from an empty laptop.

YOU WILL NEED	DETAILS
A laptop	Windows, macOS or Linux, with permission to install programs and about 5 GB of free disk
A Google account	Any Gmail address works
A bank card	Google Cloud asks for one to open a billing account, even for small use
About an hour	For sections 3 to 5, which you only ever do once. Docker is the long one
About 90 minutes	For the eight steps

What it costs. Cloud Run only charges while it is answering requests, and it goes down to zero instances when nobody is using it. The load tests in this guide send short chat lines with no searches. Running the whole guide once, including every load test, costs well under a dollar. Section 17 shows how to delete everything afterwards so nothing is left running.

2.1 How to open a terminal

A terminal is a window where you type commands. You will use it throughout.

ON	DO THIS
Windows	Press the Windows key, type **PowerShell**, press Enter
macOS	Press Cmd+Space, type **Terminal**, press Enter
Linux	Press Ctrl+Alt+T

2.2 How to read the dark boxes

Dark boxes show a terminal. A bright line after the green **>** is something **you type**, followed by Enter. Do not type the **>** itself. Grey lines are what the computer answers; yours may differ slightly.

```
> python --version
Python 3.13.0
```

Type one line at a time and wait for it to finish before typing the next. Every command in this guide is a single line, even when the page wraps it onto two.

2.3 Your Project ID

Many commands contain `YOUR_PROJECT_ID`. Replace it with the Project ID you create in section 4.1, for example `my-atlas-123456`. Keep it in a note so you can paste it.

3 Install four programs

Python runs two small scripts on your laptop. Git downloads the code. The Google Cloud CLI talks to Google Cloud. Docker builds and runs containers. If you already have one of them, run its check and skip ahead.

3.1 Python

1. Go to **python.org/downloads** and press the yellow **Download Python** button.
2. Open the file that downloads.
3. **Windows only:** on the first screen, tick **Add python.exe to PATH** at the bottom. Then press **Install Now**.
4. **macOS:** press Continue through the installer and accept the defaults.

5. Close your terminal and open a new one. A terminal only notices new programs when it starts.

```
> python --version
Python 3.13.0
```

Any version from 3.11 up is fine. On macOS and Linux the command may be `python3`; use that wherever this guide says `python` until section 5.2.

3.2 Git

1. **Windows:** go to **git-scm.com/downloads**, download the Windows installer, open it, and press Next on every screen.
2. **macOS:** type `git --version` in the terminal. If Git is missing, a window offers to install the developer tools. Press Install.
3. Close your terminal and open a new one.

```
> git --version
git version 2.54.0.windows.1
```

3.3 The Google Cloud CLI

This gives you the `gcloud` command. You will use it to switch services on, to deploy, to change settings and to read logs.

1. Go to **cloud.google.com/sdk/docs/install** and choose your operating system.
2. **Windows:** download **GoogleCloudSDKInstaller.exe**, open it, and accept the defaults. On the last screen you can untick the boxes that offer to start a shell.
3. **macOS and Linux:** follow the steps on that page for your system. They end with an install script that adds `gcloud` to your terminal.
4. Close your terminal and open a new one.

```
> gcloud --version
Google Cloud SDK 571.0.0
```

3.4 Docker Desktop

Docker is the program that builds a container image and runs it. Docker Desktop is the version with an installer and a window.

1. Go to **docker.com/products/docker-desktop** and download the installer for your system.
2. **Windows:** open **Docker Desktop Installer.exe**. Leave **Use WSL 2 instead of Hyper-V** ticked and press OK. When it finishes it asks you to sign out or restart; do that. If Windows then says WSL needs an update, open PowerShell, run `wsl --update`, and start Docker Desktop again.
3. **macOS:** pick **Apple silicon** if your Mac has an M-series chip, or **Intel chip** otherwise (Apple menu, About This Mac, tells you). Open the file and drag Docker to Applications, then open it

from Applications.

4. **Linux:** follow **docs.docker.com/engine/install** for your distribution.
5. Open **Docker Desktop**. Accept the service agreement. You can skip signing in; no Docker account is needed for this guide.
6. Wait until the bottom left corner of the Docker window says **Engine running**.

Open a new terminal and check it:

```
> docker --version
Docker version 29.7.2, build a7dcaa6
```

Then run Docker's own test. It downloads a tiny image and runs it:

```
> docker run --rm hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete

Hello from Docker!
This message shows that your installation appears to be working correctly.
```

Docker must be open while you work. If a command answers "cannot connect to the Docker daemon" or "the docker daemon is not running", Docker Desktop is closed. Open it and wait for **Engine running**.

4 Set up Google Cloud

You need five things: a project, a billing account, your laptop signed in, four services switched on, and one database created. Each is a few clicks or one command.

4.1 Create a project

A project is a labelled box that holds everything you use on Google Cloud and the bill for it.

1. Open **console.cloud.google.com/projectcreate** and sign in with your Google account. The first time, accept the terms of service.
2. Type a project name, for example **my-atlas**.
3. Under the name box there is a line that begins **Project ID**. Write it down. It looks like `my-atlas-123456`. The ID is what matters, not the name.
4. Press **Create** and wait for the bell at the top right to show it is done.

4.2 Switch on billing

1. Open **console.cloud.google.com/billing**.
2. If you have no billing account, press **Create account** and follow the form. This is where the card goes. New accounts are usually offered free trial credit.

3. Open **console.cloud.google.com/billing/projects**. Find your project in the list. If its billing account column is empty, press the three dots at the end of the row, choose **Change billing**, and pick your billing account.

4.3 Sign in from the terminal

You sign in twice, and they are different things. The first signs in **you**, for the `gcloud` commands. The second signs in **programs on your laptop**, so the container you run in step 2 can call Gemini without a secret key.

```
> gcloud auth login
Your browser has been opened to visit: ...
You are now logged in as [you@gmail.com].
```

A browser tab opens. Choose your account and press Allow. Then tell `gcloud` which project to use:

```
> gcloud config set project YOUR_PROJECT_ID
Updated property [core/project].
```

```
> gcloud auth application-default login
Credentials saved to file: [...application_default_credentials.json]
```

The browser opens once more. Allow it again.

4.4 Switch on four services

On Google Cloud every service is off until you switch it on for your project. This session uses four.

SERVICE	WHAT IT DOES HERE
Cloud Run	Runs your container, and starts more copies when traffic grows
Artifact Registry	Stores the container image so Cloud Run can fetch it
Firestore	A database. It keeps the chats, so every copy of the app sees them
Vertex AI	Gives the agent the Gemini model

One command switches on all four. It takes about a minute.

```
> gcloud services enable run.googleapis.com artifactregistry.googleapis.com
firestore.googleapis.com aiplatform.googleapis.com
Operation "operations/acat.p2-..." finished successfully.
```

Prefer clicking? Open **console.cloud.google.com/apis/library**, search for each name in the table, open it, and press **Enable**.

It says billing must be enabled? The project from 4.1 is not linked to a billing account yet. Go back to 4.2, step 3.

4.5 Create the Firestore database

Switching Firestore on does not create a database. This does. A project gets one default database, and that is the one the app uses.

```
> gcloud firestore databases create --location=us-central1
Waiting for operation to complete...done.
```

If it answers that the database already exists, your project already has one and you can carry on.

Prefer clicking? Open **console.cloud.google.com/firestore**, press **Create a Firestore database**, keep **Standard edition** and **Native mode**, choose the region **us-central1**, and press **Create database**.

5 Get the code and check your setup

5.1 Download the code

Pick a folder you can find again, such as Documents. Then:

```
> git clone https://github.com/Kiruthika162003/atlas-on-cloud-run.git
Cloning into 'atlas-on-cloud-run'...
```

```
> cd atlas-on-cloud-run
```

Every command from here on is typed inside this folder. If you close the terminal, open a new one and `cd` back into it.

5.2 A small Python environment for the laptop scripts

The app's own packages are installed inside the container, not on your laptop. Your laptop only needs two small packages for the load test and the Firestore viewer.

```
> python -m venv .venv
```

```
WINDOWS
> .venv\Scripts\activate
```

```
MACOS AND LINUX
> source .venv/bin/activate
```

Your prompt now starts with `(.venv)`. Run the activate line again whenever you open a new terminal.

```
> pip install -r requirements-laptop.txt
Successfully installed google-cloud-firestore-... requests-...
```

Windows says running scripts is disabled? Run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, answer **Y**, and try the activate line again.

5.3 Tell the code which project is yours

```
WINDOWS
> copy .env.example .env
```

```
MACOS AND LINUX
> cp .env.example .env
```

Open the new `.env` file in any text editor (Notepad is fine) and replace `your-project-id` with your Project ID. Save it. The file has one line:

```
GOOGLE_CLOUD_PROJECT=my-atlas-123456
```

5.4 Check everything

One script tests every part of sections 3 and 4. It stops at the first problem and tells you what to type to fix it.

```
> python check.py

Checking your setup

  [ ok ]  Python 3.13
  [ ok ]  The packages are installed
  [ ok ]  The settings file points at project my-atlas-123456
  [ ok ]  Docker is running
  [ ok ]  gcloud is signed in as you@gmail.com
  [ ok ]  Programs on this computer are signed in
  [ ok ]  Cloud Run, Artifact Registry, Firestore and Vertex AI are switched on
  [ ok ]  The Firestore database answers

Everything works. You are ready to build the container.
```

A line that starts with `[ FIX ]` names the problem, and the line under it is the fix. Do what it says and run `python check.py` again until you see every line as `[ ok ]`.

6 What is in the folder

FILE OR FOLDER	WHAT IT IS
`atlas/`	The web app (the map page) and the recommendation agent behind it
`mcp_server/`	The places server: three map tools the agent calls
`description_agent/`	A second agent that describes each place from Google Search
`skills/`	A walking tour skill, one file of instructions
`Dockerfile`	The recipe for the container image
`serve.sh`	What the container runs when it starts
`requirements.txt`	The Python packages that go inside the container
`loadtest.py`	Sends many users at once, from your laptop
`sessions.py`	Shows the chats stored in Firestore, or clears them
`check.py`	The setup check you just ran

6.1 The four changes that make the app ready for the cloud

The agents were not rewritten. The app file, `atlas/app.py`, has four small changes, each marked with a comment. Open the file and find them.

CHANGE	WHAT	WHY THE CLOUD NEEDS IT
1	Chats can be kept in Firestore	Several copies of the app run at once, and they must share what users said
2	Every reply names the instance that sent it	So you can see scaling happen
3	The port comes from the `PORT` setting	Cloud Run decides the port, not the app
4	One log line per chat turn	In the cloud, logs are the only window into your app

Change 1 is the important one. It is six lines:

```
if os.getenv("ATLAS_SESSIONS") == "firestore":
    from google.adk.integrations.firestore.firestore_session_service import
FirestoreSessionService
    sessions = FirestoreSessionService()
else:
    sessions = InMemorySessionService()

runner = Runner(agent=root_agent, app_name="atlas", session_service=sessions,
auto_create_session=True)
```

The agent is handed a **session service**, the thing that remembers each chat. One setting decides whether that memory is inside the running program or in a database. You will flip that setting in step 8 without touching the code.

7 Step 1: put the app in a container

On your laptop the app works because your laptop has the right Python, the right packages and the right files. A container image packs all of that into one file, so it runs the same way on any machine, including Google's.

IMAGE	The packed app. A file. You build it once.
CONTAINER	A running copy of the image. You can start many from one image.
DOCKERFILE	The recipe Docker follows to build the image.

7.1 Read the Dockerfile

Open `Dockerfile`. It is short, and every line is a step in the recipe.

```
FROM python:3.13-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

ENV GOOGLE_GENAI_USE_VERTEXAI=TRUE \
    GOOGLE_CLOUD_LOCATION=global \
    ATLAS_MODEL=gemini-3.8-flash \
    PYTHONUNBUFFERED=1 \
    PYTHONWARNINGS=ignore

CMD ["sh", "serve.sh"]
```

LINE	WHAT IT MEANS
`FROM`	Start from a small Linux that already has Python 3.13
`WORKDIR`	Work in a folder called `/app` inside the image
`COPY requirements.txt` then `RUN pip install`	Install the packages. Doing this before copying the code means Docker reuses this slow step until the package list changes
`COPY . .`	Copy the code in. The file `.dockerignore` lists what to leave out, such as `.env`
`ENV`	Settings baked into the image: use Gemini through Vertex AI, and print logs straight away
`CMD`	What to run when a container starts

No key in the image. Nothing secret is copied in. On Cloud Run the service signs its own calls to Gemini and Firestore. On your laptop you lend the container your sign-in for a moment (step 2).

7.2 Read serve.sh

Atlas is three programs: the places server, the description agent and the web app. The container starts all three. The first two stay private inside the container, and only the web app listens on the port that Cloud Run sends traffic to.

```
python mcp_server/server.py &
python description_agent/agent.py &

exec gunicorn --chdir atlas --bind :${PORT:-8080} --workers 1 --threads 16 --timeout 0
app:app
```

`gunicorn` is a web server made for production. `--threads 16` lets one copy of the app talk to 16 users at the same moment. Remember that number; it comes back in step 7.

7.3 Build the image

The dot at the end means "use the Dockerfile in this folder". `-t atlas` names the image.

```
> docker build -t atlas .
 => [internal] load build definition from Dockerfile
 => [internal] load metadata for docker.io/library/python:3.13-slim
 => [1/5] FROM docker.io/library/python:3.13-slim
 => [2/5] WORKDIR /app
 => [3/5] COPY requirements.txt .
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt           185.4s
 => [5/5] COPY . .
 => naming to docker.io/library/atlas:latest
```

The first build took a little over three minutes here, nearly all of it installing packages. Build again after changing only code and it takes seconds, because the package step is reused.

On a Mac with an M-series chip build with `docker build --platform linux/amd64 -t atlas .` instead. Cloud Run runs images built for that kind of processor.

See the image you made:

```
> docker images atlas
IMAGE          ID             DISK USAGE   CONTENT SIZE
atlas:latest   152411d1171e        392MB         88.4MB
```

8 Step 2: run the container on your laptop

Before sending the image anywhere, prove it works. If it runs here, it will run on Cloud Run.

The command is long because the container starts with nothing: no project, no sign-in. You pass both in.

```
WINDOWS
> docker run --rm --name atlas-local -p 8080:8080 --env-file .env -e
GOOGLE_APPLICATION_CREDENTIALS=/adc.json -v
"$env:APPDATA\gcloud\application_default_credentials.json:/adc.json:ro" atlas
```

```
MACOS AND LINUX
> docker run --rm --name atlas-local -p 8080:8080 --env-file .env -e
GOOGLE_APPLICATION_CREDENTIALS=/adc.json -v
"$HOME/.config/gcloud/application_default_credentials.json:/adc.json:ro" atlas
```

PART	WHAT IT DOES
`--rm`	Throw the container away when it stops
`--name atlas-local`	A name for the running container, so you can spot it
`-p 8080:8080`	Connect port 8080 on your laptop to port 8080 in the container
`--env-file .env`	Hand it your Project ID
`-e` and `-v`	Lend it the sign-in file from section 4.3, read only. This is only for your laptop
`atlas`	The image to run

The three programs start. You are ready when you see these lines:

```
[INFO] Using worker: gthread
[INFO] Booting worker with pid: 9
Open. Anyone who can reach this address may call the tools.
INFO:     Uvicorn running on http://localhost:8001 (Press CTRL+C to quit)
INFO:     Uvicorn running on http://localhost:8002 (Press CTRL+C to quit)
```

Open **http://localhost:8080** in your browser and ask:

TYPE THIS IN THE CHAT BOX

I'm at Grand Central Market, Los Angeles. Find me food.

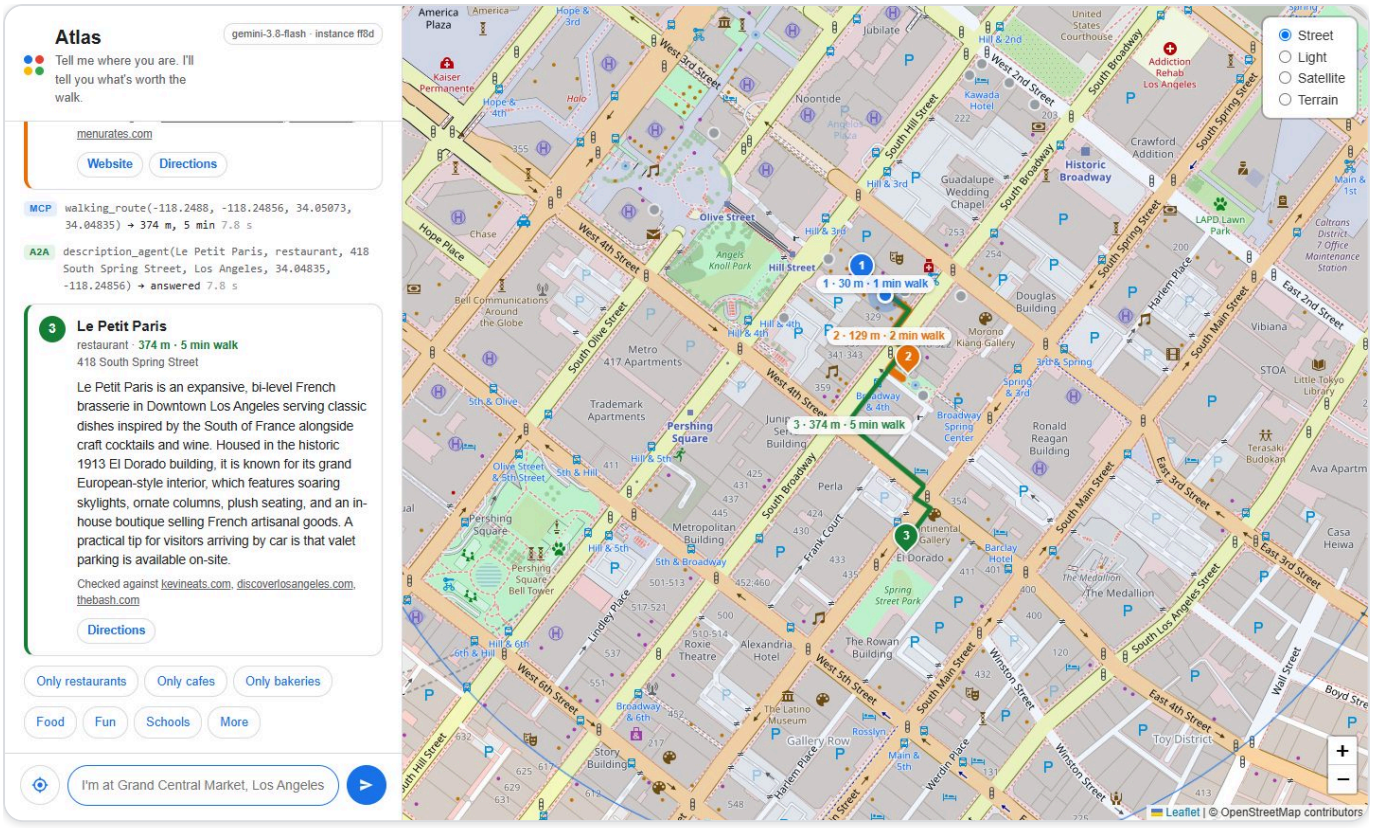


FIG 1 Atlas answering from the container on a laptop. The badge at the top says which model answered and which instance: here `ff8d`.

Look at the terminal. The app printed one line for the turn. That is Change 4:

```
turn instance=ff8d session=be91b9ff seconds=43.4 asked="I'm at Grand Central Market, Los Angeles. Find me food."
```

A full answer takes a while because it is about ten model calls and three Google searches. Open a second terminal to see the running container:

```
> docker ps
NAMES          IMAGE     STATUS              PORTS
atlas-local    atlas     Up About a minute   0.0.0.0:8080->8080/tcp
```

Go back to the first terminal and press **Ctrl+C** to stop the container.

The instance badge. Every time the app starts it gives itself a random four-character name. One running copy, one name. On Cloud Run each copy will have its own, and that is how you will see scaling.

9 Step 3: push the image to Google Cloud

The image is on your laptop. Cloud Run cannot reach your laptop, so the image has to go somewhere Google Cloud can fetch it from. That place is a registry, and Google's is called Artifact Registry.

9.1 Create a repository for images

```
> gcloud artifacts repositories create atlas --repository-format=docker --location=us-central1
Create request issued for: [atlas]
Waiting for operation [projects/my-atlas-123456/locations/us-central1/operations/...] to complete...done.
Created repository [atlas].
```

9.2 Let Docker sign in to it

```
> gcloud auth configure-docker us-central1-docker.pkg.dev
Adding credentials for: us-central1-docker.pkg.dev
Docker configuration file updated.
```

If it asks **Do you want to continue (Y/n)?**, type **Y**.

9.3 Give the image its cloud address

An image is pushed to wherever its name says. So you give your image a second name that is its full address: registry, project, repository, image and version.

```
> docker tag atlas us-central1-docker.pkg.dev/YOUR_PROJECT_ID/atlas/atlas:v1
```

It prints nothing. That is normal.

9.4 Push

```
> docker push us-central1-docker.pkg.dev/YOUR_PROJECT_ID/atlas/atlas:v1
The push refers to repository [us-central1-docker.pkg.dev/my-atlas-123456/atlas/atlas]
601b5f0ff341: Pushed
3a5db7051419: Pushed
ecc510c1e359: Pushed
9a55aeff9866: Pushed
v1: digest: sha256:0f21522798da238fc1ceefcaed3d61fd989e1f4274f77df91933e360a00b7cb5 size: 856
```

This took 72 seconds here. Each line is a layer of the image. Next time, only the layers that changed are sent.

The shortcut. Cloud Run can also do steps 1 and 3 for you: `gcloud run deploy atlas --source .` uploads the folder, builds the image in the cloud with Cloud Build, and stores it. It is handy, and it hides what you just learned. It needs one more service switched on, `cloudbuild.googleapis.com`.

10 Step 4: deploy it to Cloud Run

Cloud Run takes an image and gives you back a web address. It starts containers when requests arrive and removes them when they stop.

```
> gcloud run deploy atlas --image us-central1-docker.pkg.dev/YOUR_PROJECT_ID/atlas/atlas:v1
--region us-central1 --no-allow-unauthenticated --memory 1Gi --set-env-vars
GOOGLE_CLOUD_PROJECT=YOUR_PROJECT_ID
Deploying container to Cloud Run service [atlas] in project [my-atlas-123456] region [us-
central1]
Deploying new service...
Creating Revision.........done
Routing traffic.....done
Done.
Service [atlas] revision [atlas-00001-t4t] has been deployed and is serving 100 percent of
traffic.
Service URL: https://atlas-108279147534.us-central1.run.app
```

PART	WHAT IT DOES
`atlas`	The name of the service
`--image`	The image you just pushed
`--region`	Where in the world it runs
`--no-allow-unauthenticated`	Private: only signed-in people you allow may call it
`--memory 1Gi`	Three Python programs need more than the default 512 MB
`--set-env-vars`	The one setting the image does not carry: your project

It took 27 seconds here. Copy the **Service URL** from the last line; you need it in step 6. Each deploy or settings change makes a new **revision**, a numbered version of the service.

10.1 Prove it is private

Paste your Service URL into a browser. You get **Error: Forbidden**. That is the service doing its job: every request your agent answers costs you money, so strangers should not be able to send them. From the terminal it is the number 403:

```
> curl -s -o /dev/null -w "%{http_code}" YOUR_SERVICE_URL
403
```

On Windows PowerShell type `curl.exe` where this guide says `curl`.

10.2 Open it as yourself

This command opens a private tunnel from your laptop to the service and attaches your sign-in to every request. Leave it running in its own terminal.

```
> gcloud run services proxy atlas --region us-central1 --port 8080
Proxying to Cloud Run service [atlas] in project [my-atlas-123456] region [us-central1]
http://127.0.0.1:8080 proxies to https://atlas-xniespgdxa-uc.a.run.app
```

The first time, it asks to install a small component. Answer **Y**. Then open **http://localhost:8080** and ask a question. Same page as before, but this time the answer comes from Google Cloud.

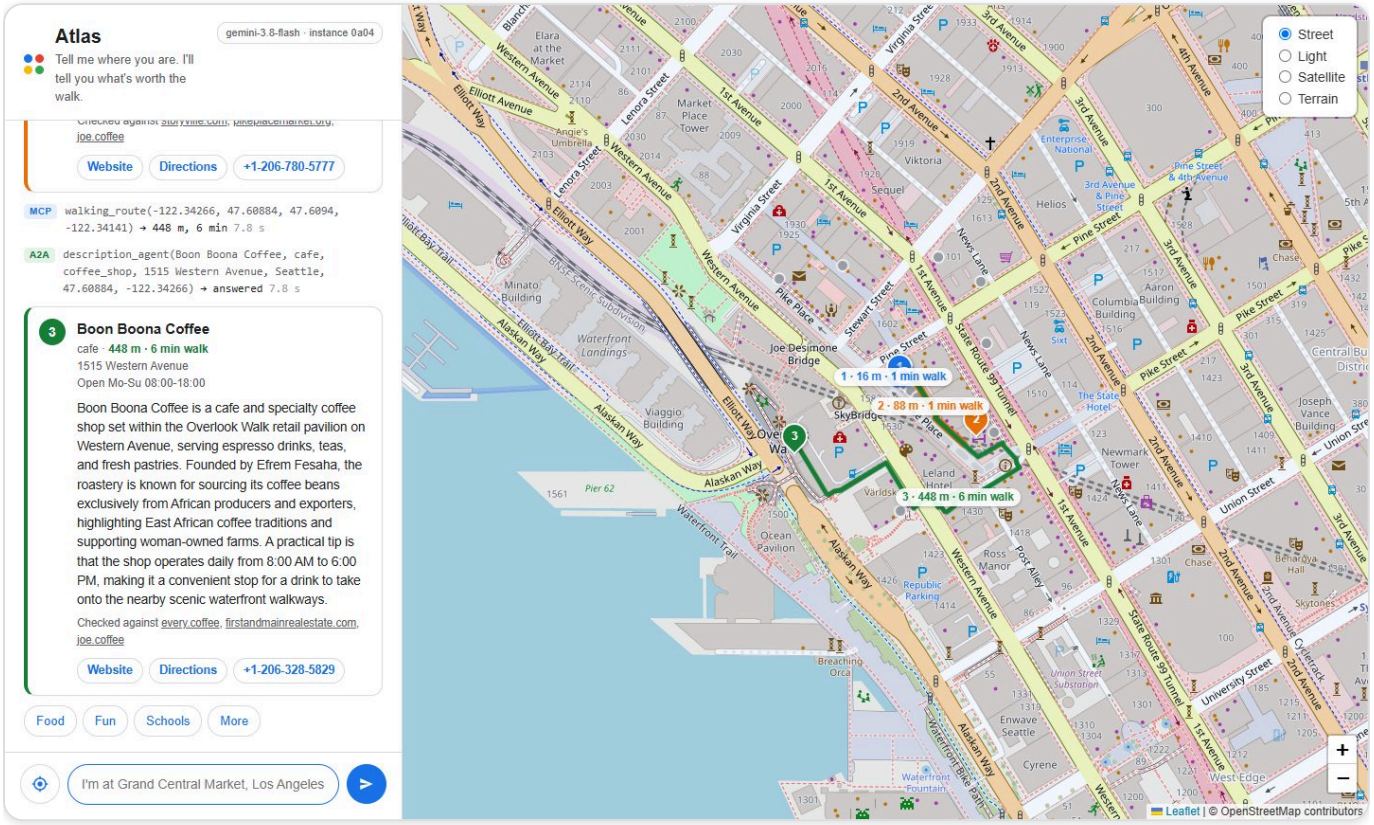


FIG 2 Atlas served from Cloud Run, asked for coffee near Pike Place Market in Seattle. The instance badge now shows a Cloud Run instance: `0a04`.

When you are done looking, press **Ctrl+C** in the proxy terminal.

> **Who signs the calls to Gemini now?** Not you. A Cloud Run service runs as a **service account**, an identity for programs. It has no password and no key file; Google Cloud vouches for it. That is why the image never needed a secret.

11 Step 5: read the logs

On your laptop you watched the terminal. In the cloud there is no terminal. Everything your app prints goes to Cloud Logging, and you read it from there.

Show only the lines the app prints for each chat turn:

```
> gcloud run services logs read atlas --region us-central1 --limit 12 --log-filter
"textPayload:turn"
2026-10-14 03:33:10 turn instance=40c8 session=bcde2948 seconds=22.6 asked='Hi, my name is
Hana.'
2026-10-14 03:33:11 turn instance=40c8 session=f5b86274 seconds=16.1 asked='What is my name?
Answer with the name only.'
2026-10-14 03:33:13 turn instance=7236 session=bcde2948 seconds=1.4 asked='What is my name?
Answer with the name only.'
2026-10-14 03:33:18 turn instance=40c8 session=85a7263e seconds=4.3 asked='What is my name?
Answer with the name only.'
```

These lines are from a later step, when the load test was running. Each one says which instance served which chat, how long it took, and what was asked. Leave off `--log-filter` to see everything, including the start-up lines of each new instance.

11.1 The same thing in the console

1. Open **console.cloud.google.com/run** and click **atlas**.
2. The **Logs** tab shows the same lines, live. Type **turn** in the filter box.
3. The **Metrics** tab has a chart called **Container instance count**. Keep this tab open during steps 6 and 7; it is where you watch instances appear.
4. The **Revisions** tab lists every version you deploy.

Write logs you can search. The line is plain `key=value` text on purpose. When something goes wrong at 2 a.m., you want to type `session=bcde2948` and see every turn of that chat, on every instance.

12 Step 6: twenty users at once

One person asking one question tells you nothing about production. The load test sends twenty people at the same moment.

Each pretend user has a two-line chat. Neither line needs a map or a search, so the whole test costs a few cents:

LINE 1	"Hi, my name is Asha."
LINE 2	"What is my name? Answer with the name only."

If the agent answers "Asha", it remembered. The script counts how many users were remembered, and which instances answered.

```
> python loadtest.py YOUR_SERVICE_URL --users 20
20 users at once, answered by 1 instance
  0a04   ######################################## 40
remembered 20 of 20 users
a two-line chat took 9.6s at best, 12.6s in the middle, 41.6s at worst
```

One instance answered all 40 lines, and everyone was remembered. So far, so good.

Why only one instance? Because Cloud Run's default is to send up to **80** requests at a time to a single instance before it starts another. Twenty is well under eighty.

Is that a good default for an agent? Not really. A web page that returns in 20 milliseconds can share an instance with 79 others. An agent turn holds a connection open for many seconds while it waits on a model. And remember `--threads 16` in `serve.sh`: this app can only work on 16 requests at once, whatever Cloud Run sends it.

13 Step 7: tune concurrency and watch it scale

Two numbers control scaling. Concurrency is how many requests one instance takes at a time. Max instances is the ceiling on how many instances there can be.

Set concurrency low, so that twenty users need more than one instance, and cap the instances at five so the bill has a ceiling too:

```
> gcloud run services update atlas --region us-central1 --concurrency 4 --max-instances 5
Done.
Service [atlas] revision [atlas-00003-jzr] has been deployed and is serving 100 percent of
traffic.
```

Nothing was rebuilt. This is a setting on the service, and it took a few seconds. Now the same load test:

```
> python loadtest.py YOUR_SERVICE_URL --users 20
20 users at once, answered by 3 instances
  5761   ################## 18
  3997   ########### 11
  0bda   ########### 11
turned away and asked again: 5 requests (HTTP 500)
remembered 6 of 20 users
a two-line chat took 19.9s at best, 38.8s in the middle, 82.9s at worst
```

Three things happened, and each one is a lesson.

13.1 It scaled

Three instances answered. Nobody started them; Cloud Run saw more requests than one instance was allowed to take and started more. This is the instance count for the whole test run, read from Cloud Monitoring:

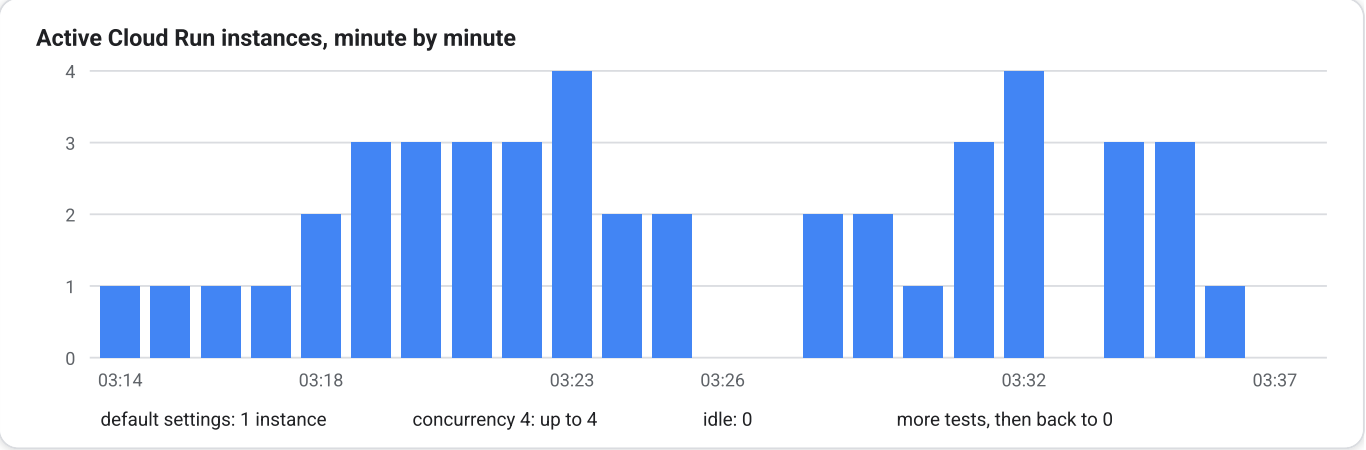


FIG 3 Real data from this guide's test runs. One instance while the default settings were on, three to four once concurrency was lowered, and zero when the tests stopped. Zero instances means zero compute cost.

In the console, the same picture is under **Metrics**, in the chart **Container instance count**.

13.2 Some requests were turned away

Five requests got an error first. Cloud Logging says why:

```
The request was aborted because there was no available instance.
```

A new instance needs a few seconds to start: fetch the image, start Python, start three programs. That is a **cold start**. During a sudden burst, a request can wait for an instance that is not up yet and be turned away with HTTP 429 or 500. The load test does what a real client should do: it waits three seconds and asks again, and counts how often it had to. Section 15 covers how to reduce this.

13.3 It forgot people

Only 6 of 20 users were remembered. Nothing crashed, and no error was shown for it. The users simply got a wrong answer. Here is why:

```
Asha:  "Hi, my name is Asha."   ──▶ instance 5761   (remembers Asha, in its own memory)
Asha:  "What is my name?"       ──▶ instance 3997   (has never heard of Asha)
```

Each instance kept its chats in its own memory. Cloud Run sends each request to whichever instance has room, so the second line of a chat often lands on an instance that never saw the first. The six who were remembered were lucky: both of their lines happened to reach the same instance.

This is the bug that only shows in production. On a laptop there is one copy of the app, so memory inside the program always works. The same code, scaled out, quietly loses conversations. An instance that is removed when traffic drops takes its chats with it too.

14 Step 8: Firestore, so chats survive

The fix is to stop keeping chats inside the instance. Put them in a database that every instance reads and writes. The app is then **stateless**: any instance can serve any user.

The code is already there; it is Change 1 from section 6.1. Switch it on with one setting. No rebuild, no push.

```
> gcloud run services update atlas --region us-central1 --update-env-vars
ATLAS_SESSIONS=firestore
Done.
Service [atlas] revision [atlas-00009-vsz] has been deployed and is serving 100 percent of
traffic.
```

Run the same load test again:

```
> python loadtest.py YOUR_SERVICE_URL --users 20
20 users at once, answered by 3 instances
  5a4d   ################ 16
  7236   ############# 13
  40c8   ########### 11
turned away and asked again: 8 requests (HTTP 429)
remembered 20 of 20 users
a two-line chat took 18.4s at best, 27.9s in the middle, 97.6s at worst
```

Still three instances. Still some requests turned away at first. But every one of the twenty was remembered, whichever instance they landed on.

SETTINGS	INSTANCES	REMEMBERED
Default (concurrency 80), chats in memory	1	20 of 20
Concurrency 4, chats in memory	3	6 of 20
Concurrency 4, chats in Firestore	3	20 of 20

14.1 See the proof in the logs

Look for one chat that was served by two different instances. The session ID is the same, the instance is not, and the answer was still right:

```
> gcloud run services logs read atlas --region us-central1 --limit 12 --log-filter
"textPayload:turn"
2026-10-14 03:33:10 turn instance=40c8 session=bcde2948 seconds=22.6 asked='Hi, my name is
Hana.'
2026-10-14 03:33:13 turn instance=7236 session=bcde2948 seconds=1.4 asked='What is my name?
Answer with the name only.'
2026-10-14 03:33:50 turn instance=7236 session=293d0ae1 seconds=53.4 asked='Hi, my name is
Bruno.'
2026-10-14 03:33:54 turn instance=5a4d session=293d0ae1 seconds=2.4 asked='What is my name?
Answer with the name only.'
```

14.2 See the chats in Firestore

```
> python sessions.py
20 chats stored in Firestore
  0b2b3223   4 events
  185baf32   4 events
  1fd0eb7b   4 events
  293d0ae1   4 events
  322f09a0   4 events
  ... and 15 more
```

Twenty chats, one per pretend user. Four events each: two things the user said and two answers.

To look at them in the console, open **console.cloud.google.com/firestore**, click the **(default)** database, and walk down this path. Each step is a click:

```
adk-session  ▶  atlas  ▶  users  ▶  visitor  ▶  sessions  ▶  (one chat)  ▶  events
```

Open an event and you can read the text the user typed and what the agent answered. That also tells you something to take seriously in a real product: chat history is now stored data. Decide who may read it and how long you keep it.

14.3 Try it yourself in the browser

1. Start the proxy again: `gcloud run services proxy atlas --region us-central1 --port 8080`
2. Open **http://localhost:8080** and ask for food near a place you know.
3. Ask a follow-up, such as "Which of those is best for vegetarians?"
4. Watch the instance badge. If it changes between your two questions and the follow-up still makes sense, you have just seen the fix work.

15 Choosing your own numbers

The settings in this guide were chosen to make scaling visible with only twenty users. For your own product, think about them like this.

SETTING	WHAT IT TRADES	A PLACE TO START FOR AN AGENT
`--concurrency`	Higher: fewer instances and a smaller bill, but slower answers when one instance is crowded. Lower: faster, more instances	No higher than the number of requests your server can really work on at once. Here that is the 16 threads in `serve.sh`
`--max-instances`	A ceiling on scale and on cost. Too low and users are turned away at busy times	Enough for your busiest expected moment, and no more. It also protects the model quota behind the agent
`--min-instances`	Keeps instances warm so nobody waits for a cold start. You pay for them all day	0 while you are building. 1 when real users are waiting
`--memory`, `--cpu`	Bigger instances start faster and take more requests each, and cost more per second	Watch the memory chart under Metrics and leave some room above the peak
`--timeout`	How long one request may run	The default is five minutes, enough for a long agent turn

Change any of them with the same command you used in step 7, for example:

```
> gcloud run services update atlas --region us-central1 --concurrency 8 --max-instances 10
```

Then run the load test again and compare. That loop, change one number and measure, is what tuning means.

15.1 What else a real launch needs

- **Clients that retry.** You saw requests turned away during a burst. Anything that calls your service should wait a moment and try again, as the load test does.
- **The model has limits too.** Cloud Run can start instances faster than your Gemini quota grows. If the logs show `429` from the model, more instances will not help.
- **Someone to let in.** The service is private. For real users you either put a sign-in in front of it or make it public on purpose with `--allow-unauthenticated`, knowing every request costs money.
- **Services you depend on.** The map data comes from free OpenStreetMap servers that ask for light use. They are fine for a workshop. A product needs a paid map service or its own copy of the data.
- **A budget alert.** Open **console.cloud.google.com/billing/budgets** and create one, so an email reaches you before a surprise does.

16 Test every part

A checklist to prove each piece to yourself. Every line is something you can run and see.

TEST	COMMAND OR ACTION	YOU SHOULD SEE
Your setup	`python check.py`	Eight lines of `[ ok ]`
Docker works	`docker run --rm hello-world`	"Hello from Docker!"
The image exists	`docker images atlas`	One row, about 390 MB
The container runs	The `docker run` line from step 2, then open localhost:8080	The map, and an answer with three places
The image is in the cloud	`gcloud artifacts docker images list us-central1-docker.pkg.dev/YOUR_PROJECT_ID/atlas`	One image named atlas
The service is up	`gcloud run services list`	A row named atlas with a URL
It is private	Open the Service URL in a browser	Error: Forbidden
It answers you	Start the proxy, open localhost:8080	The map, with a Cloud Run instance in the badge
Logs arrive	The logs command from step 5	A `turn instance=...` line for what you just asked
It scales	The load test with concurrency 4	"answered by" more than 1 instance
It remembers	The load test with Firestore on	remembered 20 of 20 users
Chats are stored	`python sessions.py`	A count of chats, each with events
Current settings	`gcloud run services describe atlas --region us-central1`	Concurrency, max instances and the env vars you set

17 Clean up, and what it costs

A Cloud Run service with no traffic runs zero instances and costs nothing for compute. The stored image and the stored chats are small, but there is no reason to leave them. Three commands remove everything this guide created.

Delete the stored chats:

```
> python sessions.py --clear
20 chats stored in Firestore
All cleared.
```

Delete the service:

```
> gcloud run services delete atlas --region us-central1
Deleting [atlas]...done.
Deleted service [atlas].
```

Delete the image repository:

```
> gcloud artifacts repositories delete atlas --location us-central1
Delete request issued for: [atlas]
Deleted repository [atlas].
```

Answer **Y** when the last two ask you to confirm. To free the disk space on your laptop as well, run `docker rmi atlas`.

WHAT YOU USED	HOW IT IS CHARGED
Cloud Run	For the seconds instances spend handling requests. There is a monthly free allowance
Gemini on Vertex AI	For the text going in and out. A full map answer is about ten model calls
Google Search grounding	Per search, on top of the model. Only full map answers use it, not the load test
Artifact Registry	For stored images, per GB per month. One 90 MB image is within the free allowance
Firestore	For reads, writes and storage. There is a daily free allowance, far above what a workshop uses
Cloud Logging	For log volume, with a large monthly free allowance

To see what you actually spent, open **console.cloud.google.com/billing**, choose your billing account, and open **Reports**. Costs can take a day to appear.

The surest way to stop all charges is to delete the project: **console.cloud.google.com/iam-admin/settings**, then **Shut down**. Only do this if the project holds nothing else you need.

18 When something goes wrong

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
"cannot connect to the Docker daemon" or "docker daemon is not running"	Docker Desktop is closed	Open it and wait for **Engine running**
Docker Desktop will not start on Windows and mentions WSL	WSL 2 is missing or old	Run `wsl --update` in PowerShell, restart, open Docker again
`docker run` says "port is already allocated"	Something else is using port 8080	Stop the other program, or use `-p 8081:8080` and open localhost:8081
The local container answers with an error about credentials	The sign-in file was not found	Run `gcloud auth application-default login`, then the `docker run` line again
`docker push` says "denied" or "unauthorized"	Docker is not signed in to the registry, or the address has a typo	Run step 9.2 again. Check the Project ID inside the image address
`docker push` says the repository does not exist	Step 9.1 was skipped, or used a different region	Run step 9.1
A command says a service "has not been used in project" or "is disabled"	That service is not switched on	Run the command from section 4.4
Deploy fails with "container failed to start and listen on the port"	The app crashed while starting	Read the logs (step 5). Run the image locally (step 2); it will fail the same way and show you why
The Service URL shows "Error: Forbidden"	Nothing is wrong. The service is private	Use the proxy (step 10.2)
The load test says every request was turned away with HTTP 403	Your sign-in was not attached	Run `gcloud auth login` and try again

The map answers "Permission denied" for Vertex AI or Firestore	The service account of the Cloud Run service lacks a role. Some organisations restrict the default	In the console open **IAM**, find the account ending `-compute@developer.gserviceaccount.com`, and add the roles **Vertex AI User** and **Cloud Datastore User**
Firestore errors mentioning that the database does not exist	Section 4.5 was skipped	Create the database, then run the test again
A row in the chat ends "the map service refused the request"	The free map service is limiting requests from that address	Wait a few seconds and ask again
Answers are slow, or the logs show `429`	The model is busy or your quota is reached	Wait a minute. Lower the number of users in the load test
On a Mac the deploy fails with "exec format error" in the logs	The image was built for the Mac's processor	Build with `--platform linux/amd64` (step 1), tag, push and deploy again

19 Small glossary

WORD	MEANING
Agent	A model with instructions and tools, working toward a goal
Image	Your app and everything it needs, packed into one file
Container	A running copy of an image
Dockerfile	The recipe for building an image
Registry	A store for images. Google's is Artifact Registry
Tag	A name and version for an image, such as `atlas:v1`
Cloud Run	A Google Cloud service that runs containers and scales them with traffic
Service	One app on Cloud Run, with one web address
Revision	A numbered version of a service. Every deploy or settings change makes one
Instance	One running copy of your container on Cloud Run
Autoscaling	Adding instances when traffic grows and removing them when it falls
Concurrency	How many requests one instance handles at the same time
Cold start	The wait while a new instance starts up
Scale to zero	No instances, and no compute cost, when there is no traffic
Stateless	An app that keeps nothing in its own memory between requests, so any copy can serve anyone
Session	One conversation: what the user said and what the agent answered
Firestore	A Google Cloud database that stores documents in collections
Service account	An identity for a program, so it can call Google Cloud without a password
Environment variable	A named setting handed to a program when it starts, such as `ATLAS_SESSIONS`
Load test	Sending many requests at once on purpose, to see what breaks
Cloud Logging	Where everything your cloud app prints ends up
HTTP 403	"You are not allowed in"
HTTP 429	"Too busy right now, try again"